

INDIAN STATISTICAL INSTITUTE

Mid-semester examination (Part-I) : Semester I (2026-2027)

BSDS Third Year

Bayesian inference

Maximum marks: 12

Time: 1 hour.

1. Assume that X is binomially distributed with parameters $n \in \mathbb{N}$ and $p \in (0, 1)$, where p is unknown. Consider a reparameterization of the binomial model using the log odds $\eta = \log \left\{ \frac{p}{1-p} \right\}$. Write the binomial model in terms of η and obtain Jeffreys' prior distribution of η directly from it. [4]

2. Let

$$X_1, \dots, X_n \mid \mu, \sigma^2 \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2),$$

where both μ and σ^2 are unknown. Assume the prior

$$\pi(\mu, \sigma^2) \propto \frac{1}{\sigma^2}.$$

- (a) Derive the marginal posterior distribution of μ .
- (b) Derive the posterior predictive distribution of a new observation $X_{n+1} \mid X_1, X_2, \dots, X_n$.
- (c) Obtain the $100(1-\alpha)\%$ equal-tailed credible interval for μ . Show that the resulting credible interval is numerically identical to the usual $100(1-\alpha)\%$ frequentist confidence interval for μ . Discuss the difference, if any, in their interpretations.

[2+2+2]

3. Construct an example in which a (95%) frequentist confidence interval, although having (95%) coverage probability over repeated samples, is guaranteed to contain the parameter for a particular set of observed data. Demonstrate this phenomenon mathematically and explain why it does not contradict the frequentist interpretation of a confidence interval. [2]

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Maximum marks: 13

Time: 1 hour.

1. Let $Y_1, Y_2, \dots, Y_n \stackrel{\text{iid}}{\sim} f(\cdot|\lambda)$ with unknown parameter $\lambda > 0$, where

$$f(y|\lambda) = \begin{cases} \exp\{-c/\lambda\} & \text{if } y = c, \\ \frac{1}{\lambda} \exp\{-y/\lambda\} & \text{if } 0 < y < c, \\ 0 & \text{otherwise.} \end{cases}$$

and $c > 0$ is a known constant.

- (a) Write down the likelihood function of λ and derive the posterior distribution under the prior, $\pi(\lambda) \propto 1/\lambda$.
- (b) Prove or disprove that the posterior is improper if all observations are equal to c .
- (c) Derive the Jeffrey's prior in this case.

[3 + 2 + 3]

2. All beds in a hospital are numbered consecutively from 1 to N (> 1). In one room a doctor sees $n \leq N$ beds, which are a random subset of all beds, with (ordered) numbers $X_{(1)} < X_{(2)} < \dots < X_{(n)}$. Consider an improper discrete uniform prior for the unknown number N of beds: $\pi(N) \propto 1$ for $N = 2, 3, \dots$. Show that for $n > 1$, the posterior probability mass function is

$$\pi(N | X_{(n)} = t) = \frac{n-1}{t} \binom{t}{n} \binom{N}{n}^{-1} \text{ for } N \geq t.$$

$$\left[\text{Hint : } \int_0^1 x^{b-1} (1-x)^{a-b} dx = \frac{1}{b} \binom{a}{b}^{-1} \right] \quad [5]$$